Internship Report

Of

WEB DEVELOPMENT

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B-Tech (CSE)

By

Roll Number of Student:- 25SCS1003002719

Name of Student: Kamlesh Kumar Yadav

Batch: 2026-27

CANDIDATE'S DECLARATION

I, **KAMLESH KUMAR YADAV**, do hereby declare that the internship report titled "**Web Development"** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: **KAMLESH KUMAR YADAV**

Roll No.: **25SCS1003002719**

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout my **Web Development Virtual Internship Program**.

First and foremost, I would like to thank **CodSoft** for providing me with the opportunity to participate in the virtual internship program. The internship provided a valuable platform to gain practical exposure to web development concepts and understand the process of developing a full-stack web application.

I am thankful to **IILM University, Greater Noida**, and the faculty members of the **School of Computer Science and Engineering** for their continuous guidance, encouragement, and support throughout my academic and professional learning journey.

I express my gratitude to my parents for their blessings.

Signature:

Student Name: **KAMLESH KUMAER YADAV**

Roll No.: **25SCS1003002719**

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3. INTERNSHIP COMPLETION CERTIFICATE

C.ID: 8a00df5

CodSoft

CERTIFICATE

OF COMPLETION

PROUDLY PRESENTED TO

Kamlesh Kumar Yadav

has successfully completed 4 weeks of a virtual internship program in

Web Development

with wonderful remarks at **CODSOFT** from **01/07/2026 to 31/07/2026.**

We were truly amazed by his/her showcased skills and invaluable contributions to the tasks and projects throughout the internship.

Founder

CERTIFIED ISO 9001-2015 COMPANY

MSME MICRO, SMALL & MEDIUM ENTERPRISES

सूक्ष्म, लघु एवं मध्यम उद्यम

contact@codsoft.in

www.codsoft.in

Date: **27/08/2026**

Internship Offer Letter

CodSoft
Code.Create.Succeed

INTERNSHIP OFFER LETTER

Date : **29/06/2026** **ID:BY26RY214181**

Dear,

Kamlesh Kumar Yadav

We would like to congratulate you on being selected for the **"Web Development"** virtual internship position with "**CodSoft**". We at **CodSoft** are excited that you will join our team.

The duration of the internship will be of **1 Month**, starting from **01 July 2026 to 31 July 2026.** The internship is an educational opportunity for you hence the primary focus is on learning and developing new skills and gaining hands-on knowledge. We believe that you will perform all your tasks/projects.

As an intern, we expect you to perform all assigned tasks to the best of your ability and follow any lawful and reasonable instructions provided to you.

We are confident that this internship will be a valuable experience for you, we look forward to working with you and helping you achieve your career goals.

By accepting this offer, you commit to executing assigned tasks diligently and ensuring excellence in all aspects of your work.

Best of Luck!

Thank You!

Kumarsh

Founder

CODSOFT CodSoft IT SERVICE

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4. PROJECT DESCRIPTION

4.1 Introduction

The primary objective of the internship was to gain practical experience in developing modern web applications and to understand how different components of a full-stack application work together. The internship provided an opportunity to strengthen my knowledge of frontend development, backend development, database operations, APIs, debugging, and version control.

As part of the internship learning experience, I worked on a project titled **“MERN Employee Dashboard.”** The project focuses on creating a centralized web-based application for managing employee-related information and operations.

The application follows a full-stack architecture using the **MERN technology stack**, consisting of React for the frontend and Node.js with Express.js for backend development, along with database integration.

The project helped me understand how a modern web application is developed from the user interface to the backend server and database. It also provided practical exposure to concepts such as **CRUD operations, REST APIs, authentication, employee management, search and filtering, and dashboard functionality**.

4.2 Organization Profile

CodSoft provides opportunities for learners to gain practical exposure and develop technical skills through internship programs and project-based learning experiences.

The Web Development Virtual Internship provided an opportunity to work on practical web development tasks and strengthen understanding of application development. The internship focused on applying programming and development concepts to build functional projects.

The internship environment encouraged self-learning, practical implementation, debugging, and problem-solving. Through the internship, I was able to explore different aspects of web application development and improve my understanding of how frontend and backend technologies work together.

The internship was especially relevant to my academic background in **Computer Science and Engineering**, as web development and full-stack application development are important areas within modern software engineering.

4.3 Problem Statement

Managing employee information manually or through multiple disconnected systems can create several difficulties. Employee details may be stored in different files, spreadsheets, or applications, making information difficult to organize and access efficiently.

Such a system can lead to problems including:

- Difficulty in maintaining centralized employee records
- Time-consuming employee information management
- Repetitive manual data entry
- Difficulty in updating existing employee information
- Lack of a structured interface for viewing employee data
- Difficulty in searching and filtering employee records
- Increased possibility of data inconsistency

Organizations require a centralized and organized system through which employee information can be managed efficiently.

The problem addressed by the **MERN Employee Dashboard** project is the need for a centralized web application that provides a structured interface for employee management and allows users to perform common employee-related operations through a single dashboard.

4.4 Project Objectives The major objectives of the project are as follows:

- To gain practical experience in full-stack web development.
- To build a centralized employee management dashboard.
- To understand the development of a React-based frontend.
- To develop backend functionality using Node.js and Express.js.
- To understand REST API communication between frontend and backend.
- To implement employee-related CRUD operations.
- To work with database operations for storing and managing application data.
- To implement authentication and secure access where applicable.
- To improve debugging and problem-solving skills.
- To gain practical experience with Git and GitHub for project management and version control.

4.5 Scope of the Project

The scope of the **MERN Employee Dashboard** project is focused on developing a centralized web application for managing employee information.

The project provides a structured interface through which employee-related operations can be performed. The application demonstrates the interaction between the frontend, backend APIs, and database.

The main scope of the project includes:

- Centralized employee information management
- Employee record creation
- Viewing employee information
- Editing existing employee records
- Deleting employee records
- Dashboard-based application interface
- Search and filtering functionality
- Backend API integration
- Database operations
- Authentication functionality

The project is designed primarily as a learning and practical implementation of full-stack web development concepts. It demonstrates how different technologies can work together to create a complete web application.

4.6 Technologies and Tools Used

The project was developed using modern web development technologies and tools.

Frontend Technologies

- React
- JavaScript
- HTML
- CSS

React was used for developing the user interface and creating a component-based frontend application.

Backend Technologies

- Node.js
- Express.js

Node.js and Express.js were used to create the backend server and develop API endpoints for communication with the frontend.

Database

- Database integration used by the project for storing and managing application data.

Development Tools

- Visual Studio Code
- Git
- GitHub
- Web Browser Developer Tools

Other Concepts Used

- REST APIs
- CRUD Operations
- Authentication
- Frontend–Backend Integration
- Database Operations
- Debugging

4.7 System Architecture The general architecture followed during the Web Development internship for building the **MERN Employee Dashboard** can be summarized in the following layers:

- **Client Layer (User):** The user accesses the application through a web browser. All interactions such as viewing, adding, updating, deleting, and searching employee information are initiated from the client side.
- **Frontend Layer (React):** The user interface is developed using React. It includes different pages and components for managing employee information and communicates with the backend through API requests.
- **Communication Layer (API Requests):** The frontend sends HTTP requests to the backend REST APIs for performing application operations. These requests are used to retrieve, create, update, and delete employee-related data.
- **Backend Layer (Node.js and Express.js):** The server-side application is developed using Node.js and Express.js. It handles incoming requests, processes application logic, and communicates with the database.
- **Routes and Controllers Layer:** Express routes define the API endpoints, while controllers handle the logic for different employee-related operations such as creating, reading, updating, and deleting employee records.

- **Database Layer:** The database is used to store and manage employee-related information. The backend performs CRUD operations to create, retrieve, update, and delete records as required.
- **Version Control Layer:** Git and GitHub are used for maintaining the project source code and tracking changes during the development process.

4.8 Methodology

The internship followed a structured, project-based methodology combining practical learning, hands-on development, implementation, testing, and version control. The development process of the **MERN Employee Dashboard** can be summarized in the table below.

Week	Module	Description
1	Requirement Understanding & Planning	Understanding the project requirements and identifying the main purpose of the Employee Dashboard. The required functionalities, target users, employee management operations, and overall project structure were planned during this stage.
1	Frontend Developmen	Developing the user interface using React, JavaScript, HTML, and CSS. The focus was on creating dashboard components and pages for displaying and managing employee-related information.
2	Backend & API Development	Developing the server-side functionality using Node.js and Express.js. REST API endpoints were created to handle communication between the frontend and backend.
2	Employee Management & CRUD Operations	Implementing the main employee management functionality, including adding, viewing, updating, and deleting employee records through CRUD operations.
3	Database Integration	Integrating the database with the backend to store and manage employee-related information. Database operations were performed for creating, retrieving, updating, and deleting records.
3	Frontend–Backend Integration	Connecting the React frontend with backend API endpoints. API requests were used to exchange data and perform employee-related operations dynamically.
4	Testing & Debugging	Testing different parts of the application, including frontend components, backend APIs, and database operations. Errors and issues were identified and resolved through debugging.
4	Version Control & Project Management	Using Git and GitHub to manage the project source code, track changes, maintain project files, and document the development work

Each week concluded with a weekly assessment to evaluate conceptual understanding, and selected modules required submission of project documentation and assignments. In the final week, a Final Assessment Test was conducted, and the overall grade was based on performance across weekly assessments, project submissions, and the final test.

4.9 Expected Outcomes

- Ability to develop and understand a complete full-stack web application using the MERN technology stack.
- Practical understanding of frontend development using **React, JavaScript**, HTML, and CSS for building interactive user interfaces.
- Hands-on experience in developing backend functionality and **REST APIs** using Node.js and Express.js.
- Ability to implement **CRUD** operations for managing employee-related information in a centralized application.
- Practical understanding of database integration and performing operations such as creating, retrieving, updating, and deleting records.
- Experience in integrating the frontend with **backend APIs** and understanding the communication between different layers of a web application.
- Improved debugging and problem-solving skills through testing and resolving frontend, backend, API, and database-related issues.
- Practical experience with Git and GitHub for version control, source code management, and maintaining the project repository.
- A completed **MERN** Employee Dashboard project demonstrating practical knowledge of full-stack web development and **project architecture.**

4.10 Certificates of Completion and Communication Proof

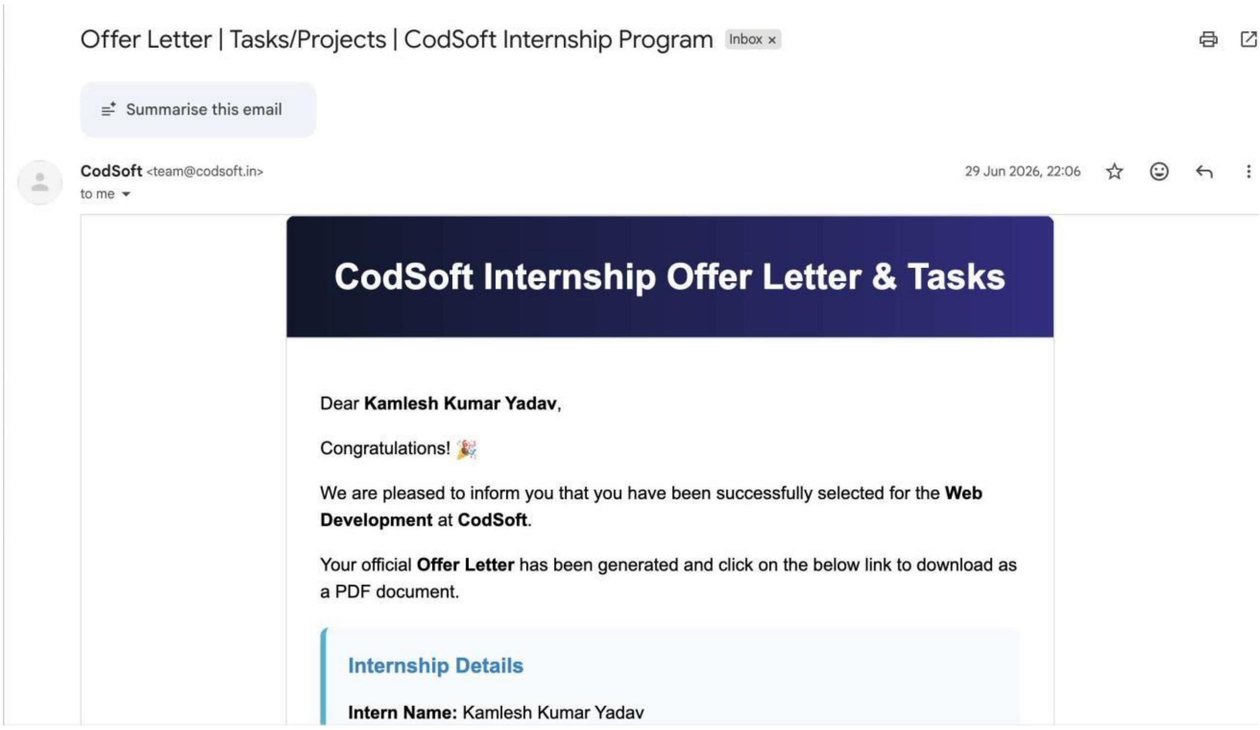
The **Internship Offer Letter** and **Internship Completion Certificate** related to the Web Development Virtual Internship at CodSoft are attached to this report.

The internship was completed during the period: **01 July 2026 – 31 July 2026**

Internship Details

1. **Internship Type:** Virtual Internship
2. **Duration:** 1 Month
3. **Student Name:** Kamlesh Kumar Yadav
4. **Roll No.:** 25SCS1003002719

5. **C.I.D.:** 879e8f9
6. **Offer Letter ID:** BY26RY214181



Offer Letter | Tasks/Projects | CodSoft Internship Program Inbox ×

Summarise this email

CodSoft <team@codsoft.in>
to me

29 Jun 2026, 22:06

CodSoft Internship Offer Letter & Tasks

Dear **Kamlesh Kumar Yadav**,

Congratulations! 🎉

We are pleased to inform you that you have been successfully selected for the **Web Development** at **CodSoft**.

Your official **Offer Letter** has been generated and click on the below link to download as a PDF document.

Internship Details

Intern Name: Kamlesh Kumar Yadav

5. BIBLIOGRAPHY/REFERENCES

- **CodSoft**, “Web Development Virtual Internship,” 2026.
- **MERN Employee Dashboard**, GitHub Project Repository: https://github.com/kamleshkumaryadav5951-pixel/mern-employee-dashboard
- **React Documentation**
- **Node.js Documentation**
- **Express.js Documentation**
- **Git and GitHub Documentation**
- **MDN Web Docs**